

VALLEY 21



Why is this project a best practice?

Challenge

The project addresses the difficulty of scaling low-energy and healthy-building principles beyond a single dwelling. It had to combine density or phased delivery with reliable energy performance, good indoor conditions, landscape quality, mobility and construction cost, while keeping the sustainability measures practical for many future occupants.

Solution

The response combines 60 planned low-impact homes designed around bioclimatic principles and green energy; a hospitality centre acting as the social and economic anchor of the community; and a mix of modern and traditional construction methods using local and sustainable materials. These measures were brought together through an integrated design approach and, where applicable, the Green Homes, nZEB or Passive House performance framework.

Replicability

The approach can be scaled through repeatable specifications for envelope, MEP systems, materials, commissioning, mobility and landscape. Replication is strongest when the developer applies the same measurable requirements to every phase and verifies them at handover, rather than treating certification as a one-off design exercise.

Key facts and figures

Location	Vama Buzăului, Braşov County, Romania
Building use / function	Low-impact rural residential and hospitality community
Project type (new build, refurbishment, extension)	New build; phased development
Plot area / Floor area	60 homes planned plus a hospitality/community centre
Year / phase	Model home completed and certified in September 2016; phased development

Climate zone	Cool humid continental / mountain foothill climate
Thematic focus / innovation	Bioclimatic design; local materials; green energy; rural landscape integration; community hospitality
Investor / building owner	Dulghiaş Development / Building the Future
Architect	Not publicly disclosed in the consulted sources
Other involved stakeholders	Romania Green Building Council; local material suppliers and craftspeople; hospitality operator
Contact person	Valley 21 / RoGBC
Further information	https://green-homes.org/property/valley-21/ https://www.rogbc.org/proiecte-certificate-rogbc?lang=en https://www.rogbc.org/_files/ugd/40ee3e_b5d91d7a3b71422cb98cd7a289b89c79.pdf

Valley 21 is a new build; phased development project in Vama Buzăului, Braşov County, Romania. It functions as low-impact rural residential and hospitality community. Its main best-practice contribution is A rural development model that uses hospitality, landscape and local materials to support both environmental and local economic value. The documented strategy brings together 60 planned low-impact homes designed around bioclimatic principles and green energy, a hospitality centre acting as the social and economic anchor of the community, and a mix of modern and traditional construction methods using local and sustainable materials. The model home was certified in 2016; later phase totals and performance should be verified with the developer. Public-source research was reviewed in July 2026; data that vary by phase are explicitly flagged in this sheet.

What was done differently?

1. Performance concept - 60 planned low-impact homes designed around bioclimatic principles and green energy.
2. Integrated systems and operation - A hospitality centre acting as the social and economic anchor of the community.
3. Wider value - A mix of modern and traditional construction methods using local and sustainable materials.

What was achieved?

Impact

Environmental impact: The project reduces energy and associated emissions primarily by lowering demand and then using efficient or renewable systems. Its material, landscape, waste or mobility measures add benefits beyond operational energy where documented. A verified whole-life carbon assessment was not publicly available, so the impact is described qualitatively unless a source provides a specific figure.

Economic impact: Lower energy demand can reduce utility exposure and improve long-term operating-cost predictability. Standardised high-performance specifications and third-party certification can also support quality assurance, market value and access to green-finance products. Public sources do not provide a complete investment, payback or life-cycle-cost analysis for this project.

Social impact: The main social benefits are stable indoor temperatures, better air quality, daylight and acoustic comfort, together with safer and more attractive shared or outdoor spaces where these form part of the project. The project also raises market awareness by making high-performance housing visible to residents, designers, contractors and investors.

Key Learnings

Success factors: Key success factors are an integrated design process, a clear performance framework, early coordination of envelope and building services, selection of competent suppliers, quality control during construction and commissioning. Transparent documentation and communication with occupants are essential to preserve the intended performance in use.

Lessons for replication: Replication should start with an energy and comfort target, not a catalogue of technologies. Design teams should verify thermal bridges, airtightness, ventilation, controls and renewable sizing together; record assumptions by project phase; commission systems; and monitor early operation. Where public figures differ by phase, the final as-built dataset must become the single source of truth.